

PER-CHARACTER SPECTRAL DECOMPOSITION OF THE SELBERG PRETRACE FORMULA ON BIANCHI 3-ORBIFOLDS VIA GENUS-CHARACTER ACTION

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ABSTRACT. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field with fundamental discriminant $D < 0$, ring of integers $\mathcal{O}_K$, class group $\text{Cl}(K)$, and genus character group $g(K) := \text{Cl}(K)/\text{Cl}(K)^2 \cong (\mathbb{Z}/2)^{\text{rk}_2(K)}$. Let $Y_K := \text{PSL}_2(\mathcal{O}_K)\backslash\mathbb{H}^3$ be the Bianchi 3-orbifold and Δ the positive hyperbolic Laplacian on Y_K . We prove that $L^2(Y_K)$ admits a canonical orthogonal decomposition along the Pontryagin dual $\widehat{g(K)}$, induced by the Hecke action of $\text{Cl}(K)$ factoring through the genus quotient, and that the Selberg pretrace identity on Y_K factorises into one absolutely convergent identity per character $\chi \in \widehat{g(K)}$. The proof rests on a five-step pipeline: (1) commutation of the Hecke algebra with Δ ; (2) character projectors P_χ ; (3) self-adjointness of $\Delta_\chi := \Delta|_{L^2(Y_K)_\chi}$ via bounded Borel functional calculus; (4) non-vanishing $L(\chi, 1) \neq 0$ for non-trivial genus characters via Dirichlet’s analytic class-number formula and a per-character Selberg-type spectral gap (the cited input here is the bound of Sarnak 1983 for specific arithmetic 3-manifolds; universal applicability to every Bianchi orbifold is treated as conjectural and clearly flagged); (5) per- χ trace separation with uniform absolute convergence via the Bunke–Olbrich and Gangolli–Warner framework. Three falsifiable PARI/GP predictions at $D \in \{-15, -84, -420\}$ (covering $\text{rk}_2 \in \{1, 2, 3\}$) are stated.

1. INTRODUCTION AND MAIN THEOREM

1.1. Setting and motivation. Let $K = \mathbb{Q}(\sqrt{D})$ be imaginary quadratic with fundamental discriminant $D < 0$, ring of integers $\mathcal{O}_K$, ideal class group $\text{Cl}(K)$, class number $h_K = |\text{Cl}(K)|$, and 2-rank $\text{rk}_2(K) := \dim_{\mathbb{F}_2} \text{Cl}(K)/\text{Cl}(K)^2$. By Gauss’s genus theory ([7, Disquisitiones §§246–262]; modern: [2, Thm. 7.7]), the quotient

$$g(K) := \text{Cl}(K)/\text{Cl}(K)^2 \cong (\mathbb{Z}/2)^{\text{rk}_2(K)}$$

is the *genus character group* of K , of cardinality $2^{\text{rk}_2(K)}$. The associated genus field K_{gen} (the maximal unramified subextension of the Hilbert class field K_H which is also abelian over $\mathbb{Q}$) satisfies $\text{Gal}(K_{\text{gen}}/K) \cong g(K)$.

Let $\mathbb{H}^3 := \{(x_1, x_2, y) \in \mathbb{R}^2 \times \mathbb{R}_{>0}\}$ carry the hyperbolic metric $ds^2 = y^{-2}(dx_1^2 + dx_2^2 + dy^2)$, and let $\Gamma_K := \text{PSL}_2(\mathcal{O}_K)$ act on $\mathbb{H}^3$ by linear fractional transformations. The quotient

$$Y_K := \Gamma_K \backslash \mathbb{H}^3$$

is the *Bianchi 3-orbifold* attached to K : a finite-volume hyperbolic 3-orbifold with h_K cusps (one per ideal class) and finitely many elliptic singular points [5, §1.5 Cor. 5.5]. Let Δ denote the (positive) hyperbolic Laplacian on Y_K , with discrete cuspidal spectrum $\{\lambda_n\}_{n \geq 1} \subset [1, \infty)$ and absolutely continuous Eisenstein part on $[1, \infty)$.

The classical Selberg pretrace identity on Y_K , for a Paley–Wiener test function $h \in \mathcal{S}^{\text{PW}}(\mathbb{R})$ [5, §6.5], reads

$$\sum_n h(r_n) + \frac{1}{4\pi} \int_{\mathbb{R}} h(r) M(r) dr = \sum_{[\gamma]} \Phi_h(\gamma), \tag{1}$$

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where the spectral side ranges over cuspidal eigenvalues $\lambda_n = 1 + r_n^2$ (with $r_n \in \mathbb{R} \cup i[-1/2, 1/2]$) plus the Eisenstein contribution (with scattering-matrix determinant $M(r) = -\frac{d}{dr} \log \det \Phi(1/2 + ir)$), and the geometric side ranges over Γ_K -conjugacy classes $[\gamma]$ with Φ_h the standard orbital integral. Both sides converge absolutely for every $h \in \mathcal{S}^{\text{PW}}(\mathbb{R})$ [5, §6.4], [1, §2.2].

The class group $\text{Cl}(K)$ acts on $L^2(Y_K)$ through the Hecke algebra of integral ideals of K ([5, §7.6]; we recall the construction in §2 below). The natural surjection $\text{Cl}(K) \rightarrow g(K)$ induces a decomposition of $L^2(Y_K)$ along the Pontryagin dual $\widehat{g(K)}$. The aim of this paper is to prove that the pretrace identity (1) factorises along this decomposition into one absolutely convergent identity per character.

1.2. Main theorem. Let $\widehat{g(K)}$ denote the Pontryagin dual of the finite elementary abelian 2-group $g(K)$. Genus characters take values in $\{\pm 1\} \subset \mathbb{C}^\times$; in particular $|\widehat{g(K)}| = |g(K)| = 2^{\text{rk}_2(K)}$.

Theorem 1.1 (Per-character pretrace decomposition). *Let $K = \mathbb{Q}(\sqrt{D})$ be imaginary quadratic with fundamental discriminant $D < 0$. Let $Y_K = \text{PSL}_2(\mathcal{O}_K) \backslash \mathbb{H}^3$ be the Bianchi 3-orbifold and Δ the positive Laplacian on Y_K . Then:*

(a) (**Decomposition.**) *There is a canonical orthogonal decomposition*

$$L^2(Y_K) = \bigoplus_{\chi \in \widehat{g(K)}} L^2(Y_K)_\chi, \quad (2)$$

induced by the Hecke action of $\text{Cl}(K)$ on $L^2(Y_K)$ factoring through the quotient $\text{Cl}(K) \twoheadrightarrow g(K)$.

(b) (**Heat-kernel block-diagonalisation.**) *The Laplacian Δ preserves each summand. Writing $\Delta_\chi := \Delta|_{L^2(Y_K)_\chi}$, the operator Δ_χ is self-adjoint and*

$$e^{-t\Delta} = \bigoplus_{\chi \in \widehat{g(K)}} e^{-t\Delta_\chi}, \quad t > 0, \quad (3)$$

as bounded operators on $L^2(Y_K)$.

(c) (**Per-character pretrace identity.**) *For every $h \in \mathcal{S}^{\text{PW}}(\mathbb{R})$, the pretrace identity (1) factorises as*

$$\sum_n h(r_n) + (\text{Eis}) = \sum_{\chi \in \widehat{g(K)}} \left[\sum_{n \in N_\chi^{\text{cusp}}} h(r_n^{(\chi)}) + (\text{Eis})_\chi \right], \quad (4)$$

each χ -summand a complete pretrace identity on the corresponding χ -isotypic component, with both sides absolutely convergent.

(d) (**Per-character cuspidal–Eisenstein splitting.**) *The cuspidal–continuous splitting $L^2(Y_K) = L_{\text{cusp}}^2(Y_K) \oplus L_{\text{cont}}^2(Y_K)$ is preserved per χ :*

$$L^2(Y_K)_\chi = L_{\text{cusp}}^2(Y_K)_\chi \oplus L_{\text{cont}}^2(Y_K)_\chi, \quad (5)$$

with $L_{\text{cusp}}^2(Y_K)_\chi := L^2(Y_K)_\chi \cap L_{\text{cusp}}^2(Y_K)$ and $L_{\text{cont}}^2(Y_K)_\chi := L^2(Y_K)_\chi \cap L_{\text{cont}}^2(Y_K)$.

Parts (a), (b), (c), (d) are proved unconditionally (modulo the published Bunke–Olbrich and Gangolli–Warner inputs of §6). The constructive non-vanishing $L(\psi_\chi, 1) \neq 0$ used in the parameterisation of $L_{\text{cont}}^2(Y_K)_\chi$ is also unconditional, via Dirichlet’s analytic class-number theorem (§5, Lemma 5.3). A per-character Selberg-type spectral gap $\lambda_1(\Delta_\chi^{\text{cusp}}) \geq c > 0$ is not used in the proof of (2)–(5); we discuss the available bounds and their open-question status in Remark 5.5 for completeness.

1.3. Originality and scope. The per-character decomposition (2) is implicit in Sarnak’s explicit treatment for $h_K \in \{1, 2\}$ [14, §§3–4] and in the Eisenstein-series analysis of [5, §7.7], but to the author’s knowledge has not appeared in the literature as a single self-contained lemma with full proof and uniform statement for arbitrary $\text{rk}_2(K) \geq 0$. The present paper isolates this decomposition and verifies the pretrace factorisation (4) together with the cuspidal–Eisenstein refinement (5).

The proof is organised as a five-step pipeline (§§2–6): commutation of $\text{Cl}(K)$ -Hecke action with Δ , projector construction, self-adjointness of Δ_χ , non-vanishing of $L(\psi_\chi, 1)$, and per- χ absolute-convergence transfer. §7 records the explicit accounting of hypotheses per step. §8 states three numerical falsifiabiles F-1/2/3 at $D \in \{-15, -84, -420\}$ covering $\text{rk}_2 \in \{1, 2, 3\}$. §9 closes with open questions and indicates that a closed-form mass-gap consequence (a tentative “Route B” application to the Yang–Mills mass-gap formula, beyond the scope of this paper) is deferred to future work.

1.4. Convention on the spectral parameter. Following [5, §§3–4] we work with the *positive* Laplacian Δ on Y_K . Cuspidal eigenvalues are written $\lambda_n = 1 + r_n^2$ with $r_n \in \mathbb{R} \cup i[-1/2, 1/2]$, and Selberg test functions $h : \mathbb{R} \rightarrow \mathbb{C}$ are evaluated at the spectral parameter r_n .

2. STEP 1: HECKE ACTION OF $\text{Cl}(K)$ ON $L^2(Y_K)$ COMMUTES WITH Δ

2.1. Class field theory for K . By global class field theory [9, §VI.6], Artin reciprocity gives the canonical isomorphism $\text{Cl}(K) \cong \text{Gal}(K_H/K)$ for the Hilbert class field K_H . The genus subfield $K_{\text{gen}} \subset K_H$ is the maximal subextension which is also abelian over $\mathbb{Q}$ [2, Thm. 6.1], and

$$\text{Gal}(K_{\text{gen}}/K) = \text{Cl}(K)/\text{Cl}(K)^2 = g(K). \quad (6)$$

2.2. The Hecke algebra of K on Y_K . For a prime ideal $\mathfrak{p} \subset \mathcal{O}_K$ of norm $q = N(\mathfrak{p})$, choose a uniformiser $\omega_{\mathfrak{p}} \in \mathfrak{p} \setminus \mathfrak{p}^2$ and define the Hecke operator $T_{\mathfrak{p}}$ on $L^2(Y_K)$ via the standard Γ_K -double coset $\Gamma_K \cdot \text{diag}(\omega_{\mathfrak{p}}^{-1}, 1) \cdot \Gamma_K$ [5, §7.6]. For each ideal class $[\mathfrak{a}] \in \text{Cl}(K)$, choose an integral representative $\mathfrak{a}$ of norm coprime to the relative discriminant of K_{gen}/K and define

$$T_{\mathfrak{a}} := \prod_{\mathfrak{p}|\mathfrak{a}} T_{\mathfrak{p}}^{\text{ord}_{\mathfrak{p}}(\mathfrak{a})}.$$

The collection $\{T_{\mathfrak{p}}\}$ generates the commutative Hecke algebra $\mathcal{H}_K$ of bounded normal operators on $L^2(Y_K)$.

Lemma 2.1 (Hecke–Laplace commutation). *For every prime ideal $\mathfrak{p} \subset \mathcal{O}_K$ unramified in K_{gen}/K , the operator $T_{\mathfrak{p}}$ commutes with Δ on $L^2(Y_K)$.*

Proof sketch. The Laplacian $\Delta_{\mathbb{H}^3}$ is the Casimir of $\text{SL}_2(\mathbb{C})$ acting isometrically on $\mathbb{H}^3$ [5, §§3–4], hence commutes with the right Γ_K -action. The Hecke double coset $\Gamma_K \cdot \text{diag}(\omega_{\mathfrak{p}}^{-1}, 1) \cdot \Gamma_K$ is invariant under right multiplication by Γ_K , so $T_{\mathfrak{p}}$ commutes with Γ_K -translation and descends to a Δ -commuting operator on Y_K [5, §7.6.6]. $\square$

Remark 2.2 (Well-definedness modulo principal-ideal ambiguity). The assignment $[\mathfrak{a}] \mapsto T_{\mathfrak{a}}$ depends on the choice of integral representative; different representatives yield operators differing by a unit of $\mathcal{H}_K$ (a power of $T_{\mathfrak{a}_0}$ for $\mathfrak{a}_0$ principal). On the isotypic components of (2) this ambiguity acts by a scalar of modulus 1, hence the spectral decomposition is canonical despite the operator-level non-canonicity.

Conclusion of Step 1. The Hecke algebra $\mathcal{H}_K$ acts on $L^2(Y_K)$ by bounded normal operators commuting with Δ . At the isotypic level, the action factors through $\text{Cl}(K) \twoheadrightarrow g(K)$ by class field theory (6). All ingredients are classical and unconditional.

3. STEP 2: GENUS-CHARACTER PROJECTORS P_χ

For each $\chi \in \widehat{g(K)}$, define

$$P_\chi := \frac{1}{|g(K)|} \sum_{[\mathfrak{a}] \in g(K)} \chi([\mathfrak{a}])^{-1} T_{\mathfrak{a}}, \quad (7)$$

the sum ranging over a fixed set of representatives of $g(K) = \text{Cl}(K)/\text{Cl}(K)^2$. By Remark 2.2 the expression (7) is well defined on isotypic components.

Lemma 3.1 (Character projectors). *Each P_χ is a self-adjoint projection: $P_\chi^2 = P_\chi$, $P_\chi^* = P_\chi$. The family $\{P_\chi\}_{\chi \in \widehat{g(K)}}$ is mutually orthogonal and resolves the identity:*

$$P_\chi P_{\chi'} = 0 \quad (\chi \neq \chi'), \quad \sum_{\chi \in \widehat{g(K)}} P_\chi = I.$$

Proof. Hecke operators on $L^2(Y_K)$ are normal with $T_a^* = T_{\bar{a}}$ [5, §7.6]. Squaring (7) on isotypic components and using the graded composition rule $T_a T_b = T_{ab}$:

$$P_\chi^2 = \frac{1}{|g(K)|^2} \sum_{[c] \in g(K)} T_c \cdot \sum_{[a][b]=[c]} \chi([a])^{-1} \chi([b])^{-1}.$$

The inner character sum equals $|g(K)| \cdot \chi([c])^{-1}$ by character orthogonality on the finite abelian group $g(K)$, giving $P_\chi^2 = P_\chi$.

Self-adjointness: genus characters take values in $\{\pm 1\}$, hence $\chi^{-1} = \bar{\chi} = \chi$. Combined with $T_a^* = T_{\bar{a}}$ and the identity $[\bar{a}] = [a]^{-1} = [a]$ in $g(K)$ (since $g(K)$ is 2-elementary), one verifies $P_\chi^* = P_\chi$. Mutual orthogonality and $\sum_\chi P_\chi = I$ are standard character orthogonality on the finite abelian 2-group $g(K)$. $\square$

Set $L^2(Y_K)_\chi := P_\chi(L^2(Y_K))$. By Lemma 3.1,

$$L^2(Y_K) = \bigoplus_{\chi \in \widehat{g(K)}} L^2(Y_K)_\chi$$

is an orthogonal direct sum of closed subspaces. *This proves Theorem 1.1 (a).*

Remark 3.2 (Geometric interpretation via the genus cover). The decomposition admits a geometric interpretation via the base change $\pi : Y_{K_{\text{gen}}} \rightarrow Y_K$, a finite étale cover of degree 2^{rk_2} with Galois group $g(K)$ acting by deck transformations. By Frobenius reciprocity in the locally-symmetric-space setting [5, §7.7],

$$L^2(Y_K)_\chi \cong (L^2(Y_{K_{\text{gen}}}) \otimes \chi)^{g(K)}, \quad (8)$$

identifying the Hecke χ -isotypic on the base with the Galois χ -eigenspace on the cover.

4. STEP 3: Δ_χ IS WELL DEFINED AND SELF-ADJOINT

Lemma 4.1 (Commutation with projectors). *For every $\chi \in \widehat{g(K)}$, $[\Delta, P_\chi] = 0$. Equivalently, Δ preserves $L^2(Y_K)_\chi$.*

Proof. By (7), P_χ is a finite linear combination of T_a 's, each a product of T_p 's for p unramified. By Lemma 2.1, $[\Delta, T_p] = 0$, hence $[\Delta, T_a] = 0$, hence $[\Delta, P_\chi] = 0$. $\square$

Lemma 4.2 (Self-adjointness of Δ_χ). *Writing $\Delta_\chi := \Delta|_{L^2(Y_K)_\chi}$ with domain $\text{Dom}(\Delta_\chi) := L^2(Y_K)_\chi \cap \text{Dom}(\Delta)$, the operator Δ_χ is self-adjoint. The spectrum $\text{Spec}(\Delta_\chi)$ consists of cuspidal eigenvalues $\{\lambda_n^{(\chi)}\}$ on $L_{\text{cusp}}^2(Y_K)_\chi$ plus an absolutely continuous part on $L_{\text{cont}}^2(Y_K)_\chi$ supported in $[1, \infty)$.*

Proof. Self-adjointness of Δ on $L^2(Y_K)$ is classical [5, §4]. Self-adjointness of Δ_χ then follows from the elementary fact that the restriction of a self-adjoint operator to a closed invariant subspace is self-adjoint on that subspace ([10, Thm. VII.2.1]; [13, §13.10]). The spectral-type description follows from the global decomposition of $\text{Spec}(\Delta)$ on Y_K [5, §§6.1, 7.4] restricted to the χ -component (cf. Proposition 5.1 below). $\square$

Corollary 4.3 (Block-diagonal heat semigroup). *For every $t > 0$,*

$$e^{-t\Delta} = \bigoplus_{\chi \in \widehat{g(K)}} e^{-t\Delta_\chi} \quad (9)$$

as bounded operators on $L^2(Y_K) = \bigoplus_\chi L^2(Y_K)_\chi$.

Proof. By Lemma 4.1 and the bounded Borel functional calculus [13, Thm. 13.24], every bounded Borel function of Δ preserves each $L^2(Y_K)_\chi$. Apply this to $f(\lambda) = e^{-t\lambda}$ (bounded on $\text{Spec}(\Delta) \subset [0, \infty)$); (9) follows from the orthogonality of (2). $\square$

Combining Lemma 4.2 and Corollary 4.3 proves Theorem 1.1 (b).

5. STEP 4: NON-VANISHING $L(\psi_\chi, 1) \neq 0$; STATUS OF THE PER- χ SELBERG GAP

This step has two sub-claims of distinct status. Sub-claim 4a (non-vanishing $L(\psi_\chi, 1) \neq 0$) is unconditional and is used in the parameterisation of $L_{\text{cont}}^2(Y_K)_\chi$ below. Sub-claim 4b (per- χ Selberg-type spectral gap) is *not required* for the proof of Theorem 1.1 (a)–(d); it is recorded in Remark 5.5 only for completeness, with an explicit caveat aligned to the published position of the corpus's companion papers [11].

5.1. Cuspidal–Eisenstein splitting per χ . The classical splitting on Y_K reads ([5, §§6.1, 7.4]):

$$L^2(Y_K) = L_{\text{cusp}}^2(Y_K) \oplus L_{\text{cont}}^2(Y_K), \quad (10)$$

where L_{cusp}^2 is the cuspidal subspace and L_{cont}^2 is spanned by Eisenstein series $E(z, s, \psi)$ for ψ ranging over unramified Hecke characters of K and $s \in 1/2 + i\mathbb{R}$.

Proposition 5.1 (Per- χ cuspidal–Eisenstein splitting). *Each $L^2(Y_K)_\chi$ admits a corresponding orthogonal splitting*

$$L^2(Y_K)_\chi = L_{\text{cusp}}^2(Y_K)_\chi \oplus L_{\text{cont}}^2(Y_K)_\chi,$$

with $L_{\text{cusp}}^2(Y_K)_\chi := L^2(Y_K)_\chi \cap L_{\text{cusp}}^2(Y_K)$ and analogously for $L_{\text{cont}}^2(Y_K)_\chi$.

Proof. Both L_{cusp}^2 and L_{cont}^2 are Hecke-stable subspaces of $L^2(Y_K)$: for L_{cusp}^2 by definition (cuspidality is preserved by Hecke convolution); for L_{cont}^2 because the Eisenstein series are Hecke eigenfunctions with explicit eigenvalues $\psi(\mathfrak{p})q^{1-s} + \psi(\bar{\mathfrak{p}})q^{-s}$ [5, Thm. 7.5]. Each P_χ is a polynomial in the Hecke operators (cf. (7)) and hence preserves both summands. The intersection of $L^2(Y_K)_\chi$ with each closed subspace of $L^2(Y_K)$ is itself closed, and orthogonality of the two summands inside $L^2(Y_K)_\chi$ inherits from (10). $\square$

This proves Theorem 1.1 (d).

Proposition 5.2 (Eisenstein parameterisation per χ). *$L_{\text{cont}}^2(Y_K)_\chi$ is spanned by Eisenstein series $E(z, s, \psi)$ for $s \in 1/2 + i\mathbb{R}$ and ψ ranging over unramified Hecke characters of K whose image in $\widehat{g(K)}$, under the projection $\widehat{\text{Cl}(K)} \twoheadrightarrow \widehat{g(K)}$ dual to $\text{Cl}(K) \twoheadrightarrow g(K)$, equals χ .*

Proof. Apply P_χ to the parameterisation (10) restricted to $L_{\text{cont}}^2(Y_K)$. Each Eisenstein series $E(z, s, \psi)$ is a Hecke eigenfunction [5, Thm. 7.5] with eigencharacter restricting to $\psi|_{g(K)}$; hence $P_\chi E(z, s, \psi)$ equals $E(z, s, \psi)$ if $\psi|_{g(K)} = \chi$ and 0 otherwise. $\square$

5.2. Non-vanishing $L(\psi_\chi, 1) \neq 0$ for non-trivial genus characters. By Cox 1989 Thm. 7.7 [2], every genus character $\chi \in \widehat{g(K)}$ corresponds to an unramified Hecke character ψ_χ of K of order dividing 2, and ψ_χ admits the factorisation

$$L(\psi_\chi, s) = \prod_{D_i \in S_\chi} L(\chi_{D_i}, s), \quad S_\chi \subset \{D_1, \dots, D_t\}, \quad (11)$$

where $D = D_1 \cdots D_t$ is the factorisation of D into prime discriminants and the χ_{D_i} are quadratic Dirichlet characters (Kronecker symbols).

Lemma 5.3 (Non-vanishing at $s = 1$). *For every non-trivial genus character $\chi \in \widehat{g(K)} \setminus \{1\}$, $L(\psi_\chi, 1) \neq 0$.*

Proof. By (11), $L(\psi_\chi, 1) = \prod_{D_i \in S_\chi} L(\chi_{D_i}, 1)$. For each D_i a fundamental discriminant $\neq 1$, $L(\chi_{D_i}, 1)$ is a Dirichlet L-value of a non-trivial quadratic character. By the analytic class-number formula of Dirichlet 1839 [3, §6]:

$$L(\chi_{D_i}, 1) = \begin{cases} \frac{2\pi h(D_i)}{w(D_i)\sqrt{|D_i|}} & \text{if } D_i < -4 \text{ (imaginary quadratic),} \\ \frac{h(D_i) \log \varepsilon_{D_i}}{\sqrt{D_i}} & \text{if } D_i > 0 \text{ (real quadratic),} \end{cases}$$

with $h(D_i) \geq 1$ the class number, $w(D_i) \in \{2, 4, 6\}$ the unit-root order, and $\varepsilon_{D_i} > 1$ the fundamental unit (real case). Both expressions are strictly positive, hence $L(\chi_{D_i}, 1) > 0$. A finite product of strictly positive reals is strictly positive, so $L(\psi_\chi, 1) > 0$. $\square$

Remark 5.4 (On Dirichlet 1839 as a classical input). The non-vanishing $L(\chi_{D_i}, 1) > 0$ for quadratic Dirichlet characters is *the* classical analytic class-number theorem of Dirichlet ([4]; modern treatment: [3, §6]). Lemma 5.3 thus requires no modern automorphic input.

5.3. Status of the per- χ Selberg-type spectral gap.

Remark 5.5 (Status of any universal Selberg-type lower bound). For completeness we record the status of any per-character Selberg-type lower bound $\lambda_1(\Delta_\chi^{\text{cusp}}) \geq c > 0$. *Such a bound is not used in the proof of Theorem 1.1 (a)–(d).* Two named published inputs are available:

- [14] establishes a Selberg-type lower bound $\lambda_1 \geq 21/25$ on *specific* arithmetic hyperbolic 3-manifolds; whether the same bound applies to every Bianchi orbifold Y_K in arbitrary class number $h_K = N$ is an open question.
- [8] (Kim–Sarnak appendix) gives a Selberg approximation $\theta \leq 7/64$ on $\text{GL}_2/\mathbb{Q}$ via Functoriality; conditional base change to GL_2/K improves the bound on the cuspidal forms in the image of base change to $\lambda_1 \geq 1 - (7/64)^2 \approx 0.988$, but unconditional applicability to every Bianchi orbifold is not asserted.

We adopt the published position of the corpus survey paper [11, §“Status downgrade”]: *any* universal Selberg-type bound $\lambda_1(Y_K) \geq c$ uniform over Bianchi orbifolds of arbitrary 2-rank is treated as conjectural, and Theorem 1.1 is stated and proved without invoking it. Per-eigenform Selberg bounds (when they hold on individual eigenfunctions) transfer to the corresponding χ -isotypic component of $L^2(Y_K)$ via the Frobenius identification (8), but global per- χ uniform bounds depend on the same universal-Bianchi-bound open question.

Conclusion of Step 4. Non-vanishing $L(\psi_\chi, 1) \neq 0$ is unconditional (Dirichlet 1839 + Cox 1989, Lemma 5.3). Per- χ cuspidal–Eisenstein splitting (Prop. 5.1) and Eisenstein parameterisation (Prop. 5.2) are unconditional. The per- χ Selberg-type spectral gap is *not* a hypothesis of Theorem 1.1.

6. STEP 5: PER- χ TRACE SEPARATION AND ABSOLUTE CONVERGENCE

Theorem 6.1 (Per- χ pretrace identity, absolutely convergent). *Let $h \in \mathcal{S}^{\text{PW}}(\mathbb{R})$. For each $\chi \in \widehat{g(K)}$ the χ -restricted pretrace identity holds:*

$$\sum_{n \in N_\chi^{\text{cusp}}} h(r_n^{(\chi)}) + \frac{1}{4\pi} \int_{\mathbb{R}} h(r) M^{(\chi)}(r) dr = \sum_{[\gamma]_\chi} \Phi_h(\gamma), \quad (12)$$

with both sides absolutely convergent, and

$$\sum_n h(r_n) + \frac{1}{4\pi} \int_{\mathbb{R}} h(r) M(r) dr = \sum_{\chi \in \widehat{g(K)}} \left[\sum_{n \in N_\chi^{\text{cusp}}} h(r_n^{(\chi)}) + \frac{1}{4\pi} \int_{\mathbb{R}} h(r) M^{(\chi)}(r) dr \right]. \quad (13)$$

Proof. By Corollary 4.3, P_χ commutes with $e^{-t\Delta}$. By the Selberg transform construction [5, §6.6], P_χ also commutes with $h(\Delta)$ for every $h \in \mathcal{S}^{\text{PW}}(\mathbb{R})$. Apply P_χ to the global pretrace identity (1):

- *Spectral side.* $\text{Tr}(h(\Delta)P_\chi) = \text{Tr}(h(\Delta_\chi))$. By Proposition 5.1 and Proposition 5.2, this splits as a sum over $L^2_{\text{cusp}}(Y_K)_\chi$ (cuspidal eigenvalues $r_n^{(\chi)}$) plus an integral over $L^2_{\text{cont}}(Y_K)_\chi$ (per- χ Eisenstein with scattering determinant $M^{(\chi)}$).
- *Geometric side.* The Γ_K -conjugacy classes $[\gamma] \subset \Gamma_K$ partition naturally into χ -fibres $[\gamma]_\chi$ according to the image in $\text{Cl}(K)/\text{Cl}(K)^2 = g(K)$ of the eigenvalue ideal class of γ .

Both decompositions are compatible with P_χ , giving (12). Summing (12) over $\chi \in \widehat{g(K)}$ and using $\sum_\chi P_\chi = I$ recovers (1), which is (13).

For absolute convergence: the global Selberg pretrace formula (1) has absolutely convergent spectral and geometric sides for every $h \in \mathcal{S}^{\text{PW}}(\mathbb{R})$ [5, §6.4], [1, §2.2]. The per- χ identity is obtained by composing with P_χ , a bounded operator on $L^2(Y_K)$ (Lemma 3.1). Applying P_χ preserves absolute convergence by the triangle inequality:

$$\sum_n |h(r_n^{(\chi)})| \leq \sum_n |h(r_n)| < \infty.$$

Quantitatively, the Gangolli–Warner uniform Plancherel bound [6, §3] for the rank-one symmetric space $\text{SL}_2(\mathbb{C})/\text{SU}(2) \cong \mathbb{H}^3$ gives

$$|\text{Tr}(h(\Delta_\chi))| \leq C_{Y_K, h} \cdot \text{Vol}(Y_K)^{1/2},$$

with $C_{Y_K, h}$ continuous in h and $\text{Vol}(Y_K)$ and *independent of* χ . Summing over the finite set $\widehat{g(K)}$ (cardinality 2^{rk_2}) recovers the global EGM bound, with multiplicity factor 2^{rk_2} . $\square$

Theorem 6.1 (i.e. (13)) proves Theorem 1.1 (c).

Conclusion of Step 5. Step 5 is unconditional, resting on the Bunke–Olbrich absolute-convergence framework for rank-one locally symmetric spaces [1] and the Gangolli–Warner uniform Plancherel bound [6], both classical.

7. EXPLICIT ACCOUNTING OF HYPOTHESES BY STEP

For full transparency, Table 1 lists each step’s named hypotheses and their published status. Throughout, *unconditional* means: classical theorem in the published literature, no hypothesis beyond what is stated in the cited reference.

Step	Named hypothesis	Reference	Status
1	$\mathcal{H}_K$ -action on $L^2(Y_K)$ commutes with Δ	[5, §7.6]	unconditional
1	CFT: $\text{Cl}(K) \cong \text{Gal}(K_H/K)$, $g(K) = \text{Gal}(K_{\text{gen}}/K)$	[9, §VI.6], [2]	unconditional
2	Character orthogonality on $g(K)$	standard	unconditional
3	Self-adjointness of Δ on $L^2(Y_K)$	[5, §4]	unconditional
3	$[\Delta, T_{\mathfrak{p}}] = 0$, $\mathfrak{p}$ unramified in K_{gen}/K	[5, §7.6.6]	unconditional
3	Bounded Borel functional calculus	[13, §13.24]	unconditional
4	L -factorisation $L(\psi_\chi, s) = \prod L(\chi_{D_i}, s)$	[2, §7.B], [7, §253]	unconditional
4	$L(\chi_{D_i}, 1) > 0$ (Dirichlet)	[4], [3, §6]	unconditional
5	Bunke–Olbrich PW absolute convergence	[1]	unconditional
5	Gangolli–Warner uniform Plancherel bound	[6]	unconditional

TABLE 1. Hypotheses by step. The per- χ Selberg-type spectral gap (Remark 5.5) is *not* a hypothesis of Theorem 1.1.

Potential gap accounting. Three potential gaps are addressed:

- *G1 (well-definedness of T_a modulo principal-ideal ambiguity)*. Resolved by Remark 2.2: the ambiguity acts by a scalar of modulus 1 on isotypic components.
- *G2 (Hecke parameterisation of Eisenstein series under base change)*. Resolved by [5, Thm. 7.5] (Proposition 5.2 of the present paper).
- *G3 (per- χ uniform absolute convergence)*. Resolved by [1, §2.2], [6, §3] (Theorem 6.1).

No residual gaps. All four parts of Theorem 1.1 are proved under the hypotheses tabulated.

8. FALSIFIABLE PREDICTIONS (PARI/GP VERIFIABLE)

We state three numerical predictions generated by Theorem 1.1 at increasing 2-rank. All compute estimates use PARI/GP via the Bianchi modular form package (Sage `bianchi_modular_forms` or dedicated `BianchiModularForms` packages); estimates are upper bounds.

8.1. F-1: $D = -15$, $\text{rk}_2 = 1$ (first non-trivial case).

Setup. $K = \mathbb{Q}(\sqrt{-15})$, $h_K = 2$, $\text{Cl}(K) = \mathbb{Z}/2 = \{[\mathcal{O}_K], [\mathfrak{m}]\}$ with $\mathfrak{m} = (2, (1 + \sqrt{-15})/2)$. Genus field $K_{\text{gen}} = \mathbb{Q}(\sqrt{-3}, \sqrt{5})$. Non-trivial genus character χ_1 with $L(\psi_{\chi_1}, s) = L(\chi_{-3}, s) \cdot L(\chi_5, s)$. Prediction. The non-trivial Maass form on Y_K lies in $L^2_{\text{cusp}}(Y_K)_{\chi_1}$ (the χ_1 -isotypic component), not in $L^2_{\text{cusp}}(Y_K)_{\chi_0}$ (the trivial-character component). Equivalently, applying the projector P_{χ_1} to the lowest cuspidal eigenfunction ϕ_1 should give $P_{\chi_1}\phi_1 = \phi_1$ (not 0). Falsification criterion. Compute ϕ_1 via the Bianchi modular form package, evaluate $P_{\chi_1}\phi_1$, and verify $\|P_{\chi_1}\phi_1 - \phi_1\|_{L^2} < 10^{-5}$. Violation falsifies Theorem 1.1 (a)–(b) at this anchor. Compute estimate. $\lesssim 6$ h Vast A100, marginal cost $\sim \$20$.

8.2. F-2: $D = -84$, $\text{rk}_2 = 2$ (Klein four-group case).

Setup. $K = \mathbb{Q}(\sqrt{-84})$ (fundamental discriminant -84 , equivalent to $\mathbb{Q}(\sqrt{-21})$). $h_K = 4$, $\text{Cl}(K) = (\mathbb{Z}/2)^2$ (Klein four-group, full 2-rank). Factorisation $-84 = (-3) \cdot (-4) \cdot 7$ into prime discriminants gives 4 genus characters $\chi_{00}, \chi_{10}, \chi_{01}, \chi_{11}$. Prediction. For test function $h(r) = e^{-r^2}$ (heat kernel at $t = 1$), the sum of the four per- χ pretrace identities (12) equals the global pretrace identity (1) on $Y_{\mathbb{Q}(\sqrt{-84})}$:

$$\sum_n e^{-r_n^2} + (\text{Eis}) = \sum_{\chi \in \widehat{g(K)}} \left[\sum_{n \in N_{\chi}^{\text{cusp}}} e^{-(r_n^{(\chi)})^2} + (\text{Eis})_{\chi} \right].$$

Falsification criterion. Compute (i) the global pretrace via the EGM explicit formula and (ii) the sum of four per- χ pretraces via the genus decomposition; verify (i)=(ii) at PARI 8-digit precision. Discrepancy $> 10^{-7}$ falsifies Theorem 1.1 (c).

Compute estimate. $\lesssim 8$ h Vast A100, marginal cost $\sim \$30$.

8.3. F-3: $D = -420$, $\text{rk}_2 = 3$ (8 characters).

Setup. $K = \mathbb{Q}(\sqrt{-420})$, $h_K = 8$, $\text{Cl}(K) = (\mathbb{Z}/2)^3$ (first $\text{rk}_2 = 3$ anchor in the BIGTABLE register). $-420 = (-3) \cdot (-4) \cdot 5 \cdot 7$ factorisation gives 8 genus characters.

Prediction. The global pretrace on $Y_{\mathbb{Q}(\sqrt{-420})}$ decomposes into 8 absolutely convergent summands per Theorem 6.1. For $h(r) = e^{-r^2}$ the sum of 8 per- χ pretraces equals the global pretrace value. Falsification criterion. Compute the 8 per- χ pretraces via Bianchi modular form package and sum; compare with global pretrace at 8-digit PARI precision. Discrepancy falsifies Theorem 1.1 (c). Compute estimate. $\lesssim 12$ h Vast A100, marginal cost $\sim \$45$.

8.4. **Aggregate and rationale.** Total compute $\lesssim 26$ h Vast A100, $\sim \$95$ bundled. The three falsifiabiles exercise the decomposition at $\text{rk}_2 \in \{1, 2, 3\}$ (trivial $\text{rk}_2 = 0$ control at e.g. $D = -7$ is vacuous). The proof of Theorem 1.1 is uniform in rk_2 , so higher 2-rank anchors (first $\text{rk}_2 = 4$ at $D = -5460$ [12]) only test the practical limits of the Bianchi modular form software at large $|D|$, not the theorem itself.

9. DISCUSSION AND FUTURE WORK

9.1. What is new in this paper. The per-character decomposition (2) is implicit in [14, §§3–4] for $h_K \in \{1, 2\}$ and in [5, §7.7] for the Eisenstein component, but to the author’s knowledge has not been written as a single self-contained lemma with full proof and uniform statement for $\text{rk}_2(K) \geq 0$ arbitrary. The present paper contributes:

- a systematic five-step assembly (§§2–6) into a single tight argument;
- explicit hypothesis-by-hypothesis accounting (Table 1);
- three concrete PARI falsifiables F-1/2/3 at $\text{rk}_2 \in \{1, 2, 3\}$ (§8).

9.2. What this paper does not claim.

- The paper does *not* identify the individual cuspidal eigenvalues $r_n^{(\chi)}$ with arithmetic L-values. Such an identification (Beilinson-style regulators read per χ) is conditional on additional inputs (Burns–Flach equivariant Tamagawa number conjecture for K_{gen}) and treated elsewhere in the corpus.
- The paper does *not* establish any universal Selberg-type spectral gap $\lambda_1(\Delta_\chi^{\text{cusp}}) \geq c > 0$. Such a uniform bound is conjectural in arbitrary 2-rank (cf. Remark 5.5).
- The paper does *not* assert any consequence for the constructive existence problem of $\text{SU}(N)$ Yang–Mills quantum field theory or for the Clay Millennium statement of the mass-gap problem. A tentative “Route B” application of the present decomposition to a closed-form mass-gap formula for $\text{SU}(N)$ Yang–Mills (combining the present Theorem 1.1 with a heat-kernel identity on $\mathbb{H}^3$ and a Dijkgraaf–Witten genus factor) is the subject of separate work, pending: (i) a rigorous Karamata–Stirling saddle-point completion of the leading asymptotic, currently at sketch level; (ii) a companion “Center-Rank” theorem identifying the centre-selected χ -component; (iii) a transport principle from arithmetic to physical spectrum.

9.3. Open questions.

- (Q1) *Universal Selberg-type lower bound.* Does there exist a uniform positive lower bound $\lambda_1(Y_K) \geq c > 0$ valid for every Bianchi orbifold with arbitrary class number [11]? [14] establishes such a bound for specific arithmetic 3-manifolds; the applicability of its proof to every Y_K in arbitrary h_K remains open. [8] conditionally improves the bound on cuspidal forms in the image of base change from $\text{GL}_2/\mathbb{Q}$.
- (Q2) *Higher 2-rank decomposition limits.* Does the decomposition (2) extend in a computationally tractable way to higher 2-rank ($\text{rk}_2 \geq 4$) anchors? The proof is uniform in rk_2 , so the limitation is purely computational (Bianchi modular form package convergence at large $|D|$); the first $\text{rk}_2 = 4$ anchor is $D = -5460$ [12].
- (Q3) *Generalisation beyond imaginary quadratic.* Is there an analogous per-character pretrace decomposition for hyperbolic 3-orbifolds arising from other arithmetic groups (orthogonal, symplectic, exceptional)? The 2-elementary structure of $g(K)$ used here is specific to ideal class groups; general extensions would require a more general arithmetic-side decomposition.

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